Tutorial-1

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A smart farming system predicts whether a plant will grow well (1) or poorly (0) based on daily sunlight hours and water amount in liters.

Dataset:

Sunlight Hours (x_1)	Water Liters (x_2)	Bias (x_3)	Target (Good Growth = 1 / poor = 0) (y)
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

① Understand and Explain the data

Inputs (x_1 & x_2) output (Target)

Mean of $x_1 = \mu x_1 = 6.6$

Mean of $x_2 = \mu x_2 = 3$

* Variance; $\boxed{\sigma^2 = \frac{1}{N} \Sigma (x_i - \mu)^2}$

where N = No. of Data

x^i = Input

μ = Mean

$$\frac{(x_i - \mu x_i)^2 + (x_i - \mu x_i)^2 + \cdots}{\text{no. of } x \text{ values}}$$

$$= \frac{(x_{11}-\mu x_1)^2+(x_{12}-\mu x_1)^2+(x_{13}-\mu x_1)^2+(x_{14}-\mu x_1)^2+(x_{15}-\mu x_1)^2}{5}$$

$$= \frac{(3-6.6)^2+(8-6.6)^2+(5-6.6)^2+(10-6.6)^2+(7-6.6)^2}{5}$$

$$= 5.84$$

$$\Rightarrow \frac{(x_i-\mu_i)^2+(x_i-\mu x_i)^2+\cdots}{\text{no. of values}}$$

$$= \frac{(x_{21}-\mu x_2)^2+(x_{22}-\mu x_2)^2+(x_{23}-\mu x_2)^2+(x_{24}-\mu x_2)^2+(x_{25}-\mu x_2)^2}{5}$$

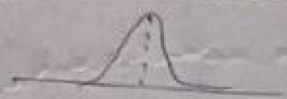
$$= \frac{(1-3)^2+(4-3)^2+(2-3)^2+(5-3)^2+(3-3)^2}{5}$$

$$= \frac{(-2)^2+(1)^2+(-1)^2+(2)^2+(0)^2}{5}$$

$$= \frac{4+1+1+4+0}{5} = \frac{10}{5} = 2.$$

variance of $x_1 \Rightarrow \sigma^2 x_1 = 5.84$

variance of $x_2 \Rightarrow \sigma^2 x_2 = 2$



$$\sigma^2 x_1 > \sigma^2 x_2$$

standard deviation of

$$\sqrt{\sigma x_1^2} = \sigma x_1 = 2.40$$

standard deviation of.

$$\sqrt{\sigma x_2^2} = \sigma x_2 \Rightarrow 1.414$$

The factor whose variance is higher is important

$\therefore$ The Sunlight Hours > water Liters

ML model

perceptron model:

It is a simple Nural Network model

It is a simple decision making model that can Answer Yes/No

perceptron:

① Inputs : x_1 & x_2

② weights associated with input : w_1 & w_2

③ Bias -> It gives the model flexibility (b)

almost it is 1

④ weighted sum : $z = w_1x_1 + w_2x_2 + b - 1$

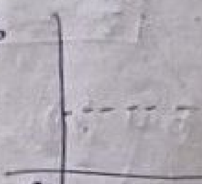
$= \sum_i w_ix_i + b$

w_1 & $w_2 = 0$

⑤ Activation function -> It is an mathematical (step function) function followed by ML

step function is a easiest function with (0 & 1s)

Activation function: $\boxed{\hat{y} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}}$



(or) 0.1

⑥ Updated weight:

$z = 0(3) + 0(1) + 1$

$w_i^{new} = w_i^{old} + \underbrace{\eta}_{eta} (y - \hat{y})x_i$

$z = 1 < 0$ 0

≥ 0 1

0.5

y = Actual Target / output

$= 0 + 1(0-1)3$

$\hat{y}$ = predicted target / output

$= 0 - 3$

$w_1 = -3$

$\boxed{\eta = \text{Learning Rate}}$

$w_2 = 0 + 1(0-1) \cdot 1$

$w_2 = -1$

prediction:

① Initial

$z = (-3)(8) + (-1)(4) + 1$

Let ; $w_1 = 0$, $w_2 = 0$, $\eta = 1$ we can take 0 - 9 also.

$= -24 - 4 + 1 < 0$ 0

Inputs = $x_1 = 3,\ x_2 = 1,\ b = 1$

$z = ?$

$z = w_1x_1 + w_2x_2 + b$

$(0)(3) + (0)(1) + 1 = 1$

$z = 1$, so check is $\hat{y} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$

$\hat{y} = 1$

but the target is "0" so we want to change the weights.

$W_1^{new} = 0 + 1(0-1) \cdot 3 = -3$

$w_2^{new} = 0 + 1(0-1) \cdot 1 = -1$

2nd case for 2nd sunlight & wales inputs

Inputs: $x_1 = 8,\ x_2 = 4,\ b = 1,\ w_1 = -3,\ w_2 = -1,\ \eta = 1,\ b = 1$

$z = (-3)(8) + (-1)(4) + 1$

$= -24 - 4 + 1$

$= -28 + 1$

$= -27$

check $\hat{y}$

$\boxed{z = -27}$; $\boxed{\hat{y} = 0}$

But in Dataset of 2nd row the target is 1 so it do not matched.

$w_1^{new} = (-3) + 1(\cancel{10} - 0)8$

$= -3 + (1-0)8$

$= -3 + 8$

$w_1^{new} = 5$

$w_2 new = -1 + 1(1-0)(4)$

$w_2 New = -1 + 4$

$w_2 New = 3$

$x_1 = 5 \quad x_2 = 2 \quad b = 1 \quad w_1 = 5 \quad w_2 = 3 \quad \eta = 1$

$z = w_1x_1 + w_2x_2 + b$

$(5)(5) + (3)(2) + 1$

$25 + 6 + 1$

$z = 32 \quad z > 0 \Rightarrow \hat{y} = 1$

$w_1new = (5) + 1(0-1)(5)$

$w_1new = 5 + (-5)$

$\boxed{w_1new = 0}$

$w_2new = (3) + 1(0-1)(2)$

$w_2new = 3-2$

$\boxed{w_2new = 1}$

$x_1 = 10, \; x_2 = 5, \; b = 1, \; w_1 = 0, \; w_2 = 1, \; \eta = 1$

$z = w_1x_1 + w_2x_2 + b$

$= (0)(10) + (1)(5) + 1$

$= 0 + 5 + 1$

$\boxed{z = 6} \quad z > 0 \text{ so } \boxed{\hat{y} = 1}$

$\therefore$ $\boxed{\text{activation function = old target}}$